Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

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***Abstract*—This document presents the mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM).**

I. Introduction

A. Problem in One Sentence

After Scout UAVs detect survivors, the Command Center must decide at each event time t which available Responder UAVs should execute which feasible tours—ordered sequences $\pi_j = (i_1, \ldots, i_k)$—to maximize expected lifesaving value under battery, payload-item, and 3D reachability constraints, before ground rescue arrives.

II. System Entities

A. Survivors

Detected tasks i, each with location $p_i \in \mathbb{R}^3$, detection time t_i^{det}, confidence $c_i \in [0,1]$, severity $\sigma_i \in [0,1]$, and demand vector $r_i^q \in \mathbb{Z}_{\geq 0}$ for $q \in \{\text{med}, \text{water}, \text{food}\}$.

B. Responder UAVs

Platforms j that execute tours, each with current position p_j, base $p_j^{\text{base}} \in \mathbb{R}^3$, battery $B_j(t) \in [0, B_{\max}]$, item capacity K, inventory I_j^q, payload $m_j = \sum_q I_j^q w_q$, and status $\in \{\text{Idle}, \text{EnRoute}\}$.

C. Command Center

Central decision unit that maintains global state $X(t)$, queries the routing oracle for each candidate tour, solves the assignment optimization, and dispatches tours π_j.

D. Environment

Disaster area $\mathcal{E} = (V, O)$ with $V \subset \mathbb{R}^3$ free space and $O \subset \mathbb{R}^3$ static obstacles (buildings, rubble), $V \cap O = \emptyset$; hazard $R(p)$ is deferred to $R = 0$ in V1.

III. System State at Time t

A. Survivor Set

$\mathcal{S}(t) = \{S_i\}$, $\mathcal{Q}_{\text{high}}(t) = \{i \in \mathcal{S}(t) \mid a_i = 0 \land c_i \geq c_{\text{thresh}}\}$ ordered by $V_i(t)$, $\mathcal{Q}_{\text{low}}(t) = \{i \mid a_i = 0 \land c_i < c_{\text{thresh}}\}$.

B. UAV Set

$\mathcal{D}(t) = \{D_j\}$, $\mathcal{D}_{\text{avail}}(t) = \{j \mid \text{status}_j = \text{Idle}\}$.

C. Global State

$X(t) = (\mathcal{S}(t), \mathcal{D}(t), \mathcal{E}, \mathcal{Q}_{\text{high}}(t), \mathcal{Q}_{\text{low}}(t))$.

IV. Survivor Priority and Value

A. Urgency

$u_i(t) = \sigma_i e^{-\alpha(t - t_i^{\text{det}})}$, $\alpha = \ln 2 / T_{\text{half}}$, $T_{\text{half}} = 30$ min, $u_i \in [0,1]$ proportional decay $du/dt = -\alpha u$, half-life, never negative.

B. Service Value

$V_i(t) = \frac{2\, u_i(t)\, c_i}{u_i(t) + c_i} \in [0,1]$ (harmonic mean, $V_i = 0$ if $u_i = 0$ or $c_i = 0$) unified expected value; $c_i \geq c_{\text{thresh}}$ hard eligibility for $\mathcal{Q}_{\text{high}}$.

C. Queue Priority

$\text{priority}(i) = \beta\, V_i(t)$, $\mathcal{Q}_{\text{high}}$ ordered by V_i descending.

V. UAV Mission and Tour

A. Tour Definition

$\pi_j(t) = (i_1, \ldots, i_k)$ with $k = \text{len}(\pi_j)$ and $\sum_{i \in \pi_j} \sum_q r_i^q \leq K$ (e.g., $K = 4$). $P_{\pi_j} = P_{p_j \to p_{i_1}} \circ \cdots \circ P_{p_{i_k} \to p_j^{\text{base}}}$.

B. Example

$K = 4$, $I_j = [\text{med}, \text{water}, \text{food}, \text{med}]$, $\pi_j = (A, C)$ with $\mathcal{R}_A = \{\text{med}\}$, $\mathcal{R}_C = \{\text{water}, \text{food}\}$ carries 3 items to 2 survivors: $\text{base}(2.8\text{kg}) \to A(10\text{s}, 3300\text{J}) \to C(5\text{s}, 1500\text{J}) \to \text{base}(12\text{s})$.

VI. Routing Module

For any path $P = \{p_0, \ldots, p_L\} \subset V$, let $T(P) = \sum_k \|p_k - p_{k-1}\| / v_{\max}$ be travel time and $E(P) = \sum_k [e_{\text{travel}}(p_k, m, w_0) \|p_k - p_{k-1}\| + e_{\text{vert}}(m) \max(0, z_k - z_{k-1})]$ be energy, where L is waypoints. A path P is feasible iff $P \subset V$.

VII. Tour Feasibility

A tour $\pi_j = (i_1, \ldots, i_k)$ is feasible iff:

- Capacity: $\sum_{i \in \pi_j} \sum_q r_i^q \leq K$,
- Inventory: $I_j^{q,(l)} = I_j^{q,(l-1)} - r_{i_l}^q$, $I_j^{q,(l)} \geq 0\ \forall q, l$,
- Battery: $E(\pi_j) + E_{\text{reserve}} \leq B_j(t)$,
- Reachability: $P_l \subset V\ \forall l$,
- Confidence: $c_i \geq c_{\text{thresh}}\ \forall i \in \pi_j$.

If any fails, $\text{Feasible}(\pi_j) = \text{false}$ and $C(\pi_j) = \infty$.

VIII. Tour Cost and Utility

For a feasible tour $\pi_j = (i_1, \ldots, i_k)$ with $T(\pi_j), E(\pi_j)$ from routing and $V_i(t)$ from §4,

$$J_{\text{tour}}(\pi_j, t) = w_T \frac{T(\pi_j)}{T_{\max}} + w_E \frac{E(\pi_j)}{E_{\max}} - w_U \frac{\sum_{i \in \pi_j} V_i(t)}{K} \quad (1)$$

where $w_T + w_E + w_U = 1$, $T_{\max} = 600\text{s}$, $E_{\max} = B_{\max} - E_{\text{reserve}}$, $V_i = 2u_i c_i/(u_i + c_i)$, K capacity. If Feasible$(\pi_j) =$ false then $J_{\text{tour}} = \infty$.

IX. Assignment Optimization

The primary assignment objective is to maximize total effective service value:

$$\max_{x_i} \sum_{i \in \mathcal{S}(t)} V_i(t)\, x_i \quad \text{s.t. } x_i \in \{0, 1\}\ \forall i \quad (2)$$

where $V_i(t) = 2u_i(t)c_i/(u_i(t) + c_i)$ is the harmonic service value from §4, $u_i(t) = \sigma_i \exp[-\alpha(t - t_i^{\text{det}})]$ with $\alpha = \ln 2/T_{\text{half}}$, and $V_i(t) \in [0, 1]$. The assignment is realized via feasible tours $\{\pi_j\}$ with $x_i = 1$ iff $i \in \bigcup_j \pi_j$, subject to payload, inventory, battery, and 3D reachability constraints from §7, and confidence threshold $c_i \geq c_{\text{thresh}}\ \forall i$ assigned to a tour (Eq. 7). Energy is a feasibility constraint ($E(\pi_j) + E_{\text{reserve}} \leq B_j$) rather than an independent optimization objective. Each survivor appears in at most one tour; unassigned survivors remain in $\mathcal{Q}_{\text{high}}$ for the next event. The tour cost $J_{\text{tour}}(\pi_j)$ from §8 is retained as an alternative weighted formulation $J_{\text{tour}} = w_T T/T_{\max} + w_E E/E_{\max} - w_U \sum V_i/K$ for $w_T + w_E + w_U = 1$.

X. System Model

XI. Problem Formulation

XII. Conclusion